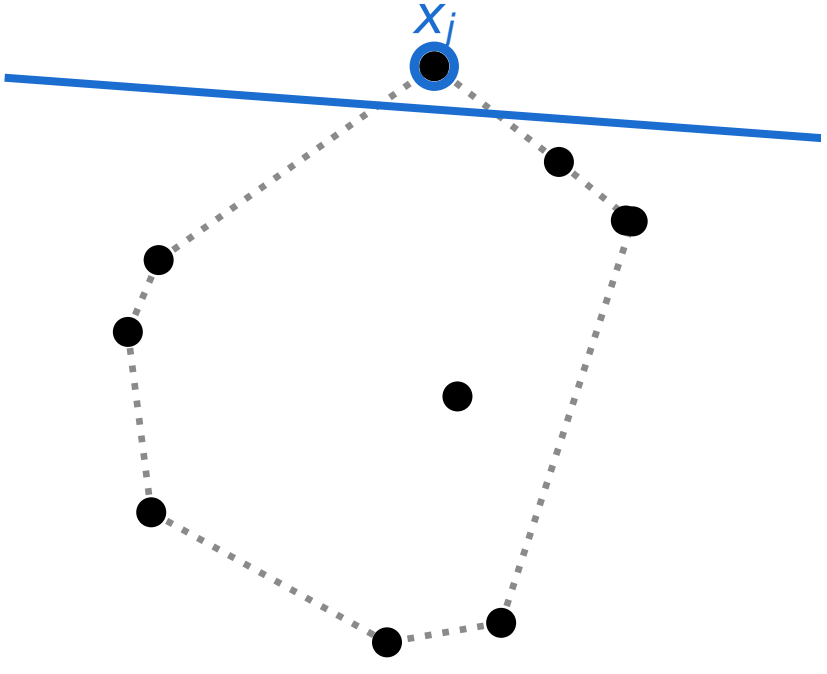
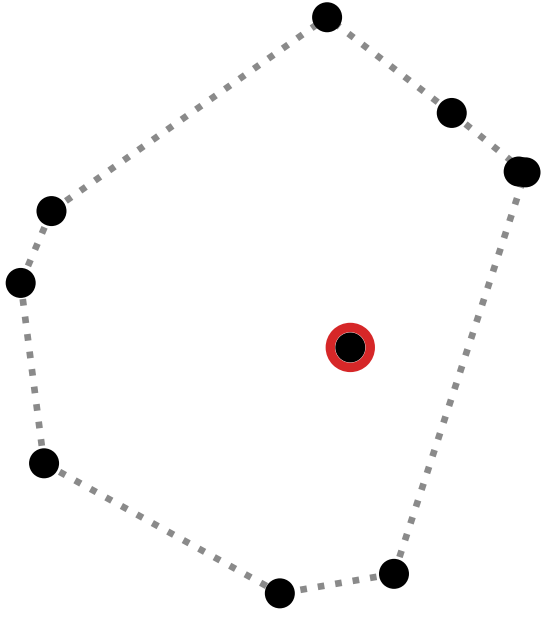
(a) A vertex can be cut off by a hyperplane



(b) An interior point cannot